

NEHA SHARMA - U24AI050 LAB-2

24 January 2026 19:22

1. Write a Shell Script to find factorial of a number.

```
#!/bin/bash
```

```
echo "Enter a number"
```

```
read n
```

```
fact=1
```

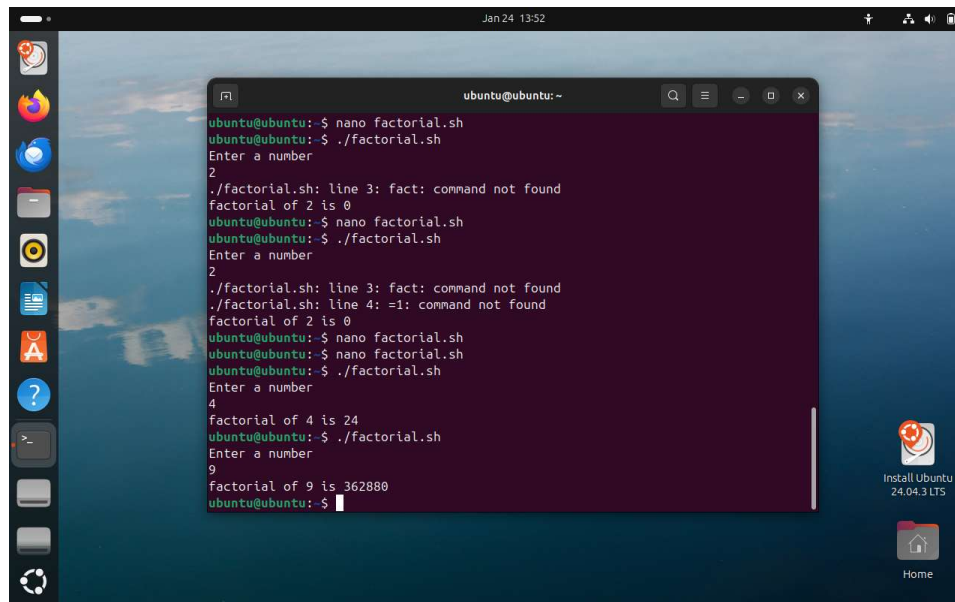
```
for ((i=1; i<=n; i++))
```

```
do
```

```
    fact=$((fact * i))
```

```
done
```

```
echo "factorial of $n is $fact"
```



The screenshot shows a terminal window on an Ubuntu desktop. The user has created a file named 'factorial.sh' and is running it multiple times. The script prompts for a number and calculates its factorial. The output shows that the factorial of 2 is 0, the factorial of 4 is 24, and the factorial of 9 is 362880. The terminal window title is 'ubuntu@ubuntu: ~'. The desktop background is a blue and white abstract image. The system clock at the top right shows 'Jan 24 13:52'. The desktop has a sidebar with application icons and a bottom panel with 'Install Ubuntu 24.04.3 LTS' and 'Home' buttons.

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ nano factorial.sh  
ubuntu@ubuntu:~$ ./factorial.sh  
Enter a number  
2  
./factorial.sh: line 3: fact: command not found  
factorial of 2 is 0  
ubuntu@ubuntu:~$ nano factorial.sh  
ubuntu@ubuntu:~$ ./factorial.sh  
Enter a number  
2  
./factorial.sh: line 3: fact: command not found  
./factorial.sh: line 4: =1: command not found  
factorial of 2 is 0  
ubuntu@ubuntu:~$ nano factorial.sh  
ubuntu@ubuntu:~$ nano factorial.sh  
ubuntu@ubuntu:~$ ./factorial.sh  
Enter a number  
4  
factorial of 4 is 24  
ubuntu@ubuntu:~$ ./factorial.sh  
Enter a number  
9  
factorial of 9 is 362880  
ubuntu@ubuntu:~$
```

2. Write a shell script for a program to implement an address book with the options given below:

- a) Create an address book.
- b) View the address book.
- c) Insert a record.
- d) Delete a record.
- e) Modify a record.
- f) Exit.

```
#!/bin/bash
```

```
file="addressbook.txt"
```

```
while true  
do
```

```
    echo "-----"  
    echo "ADDRESS BOOK MENU"  
    echo "1. Create Address Book"  
    echo "2. View Address Book"  
    echo "3. Insert a Record"  
    echo "4. Delete a Record"  
    echo "5. Modify a Record"  
    echo "6. Exit"  
    echo "-----"  
    read choice
```

```
case $choice in
```

```
1)
```

```
    touch $file  
    echo "Address book created."  
    ;;
```

```
2)
```

```
    if [ -s $file ]; then  
        cat $file  
    else  
        echo "Address book is empty."  
    fi  
    ;;
```

```
3)
```

```
    echo "Enter Name:"
```

```
read name
echo "Enter Phone:"
read phone
echo "Enter Email:"
read email
echo "$name | $phone | $email" >> $file
echo "Record added."
;;
```

4)

```
echo "Enter name to delete:"
read name
grep -v "^$name" $file > temp.txt
mv temp.txt $file
echo "Record deleted."
;;
```

5)

```
echo "Enter name to modify:"
read name
grep -v "^$name" $file > temp.txt
echo "Enter New Name:"
read newname
echo "Enter New Phone:"
read phone
echo "Enter New Email:"
read email
echo "$newname | $phone | $email" >> temp.txt
mv temp.txt $file
echo "Record modified."
;;
```

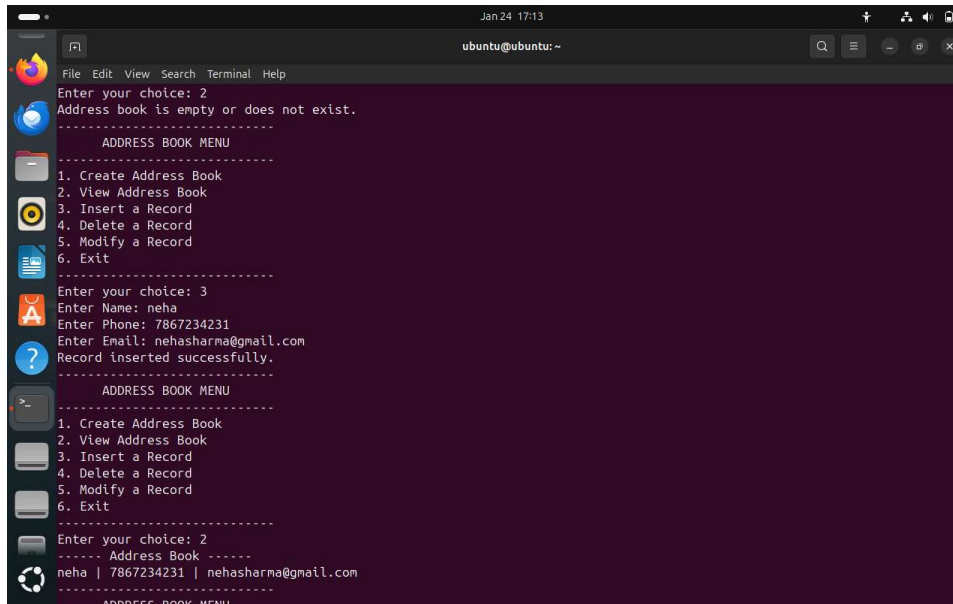
6)

```
exit
;;
```

*)

```
echo "Invalid choice"
;;
esac
```

done



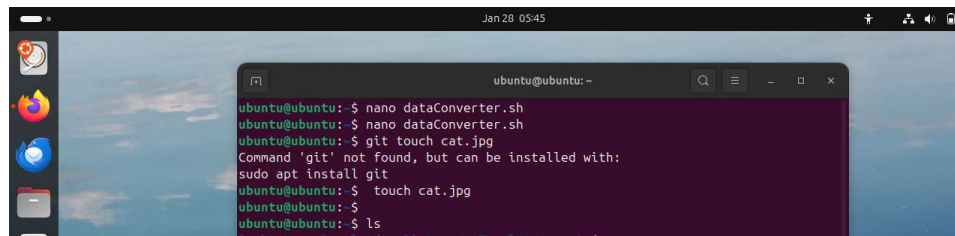
```
ubuntu@ubuntu: ~  
File Edit View Search Terminal Help  
Enter your choice: 2  
Address book is empty or does not exist.  
-----  
ADDRESS BOOK MENU  
-----  
1. Create Address Book  
2. View Address Book  
3. Insert a Record  
4. Delete a Record  
5. Modify a Record  
6. Exit  
-----  
Enter your choice: 3  
Enter Name: neha  
Enter Phone: 7867234231  
Enter Email: nehasharna@gmail.com  
Record inserted successfully.  
-----  
ADDRESS BOOK MENU  
-----  
1. Create Address Book  
2. View Address Book  
3. Insert a Record  
4. Delete a Record  
5. Modify a Record  
6. Exit  
-----  
Enter your choice: 2  
----- Address Book -----  
neha | 7867234231 | nehasharna@gmail.com  
-----  
ADDRESS BOOK MENU
```

3. Write the shell script that renames all files in the current directory that end in “.jpg” to begin with today’s date in the following format: YYYY-MM-DD.
For example, if a picture of cat was in the current directory and today was October 31,2016 it would change name from “mycat.jpg” to “2016–10–31-mycat.jpg”

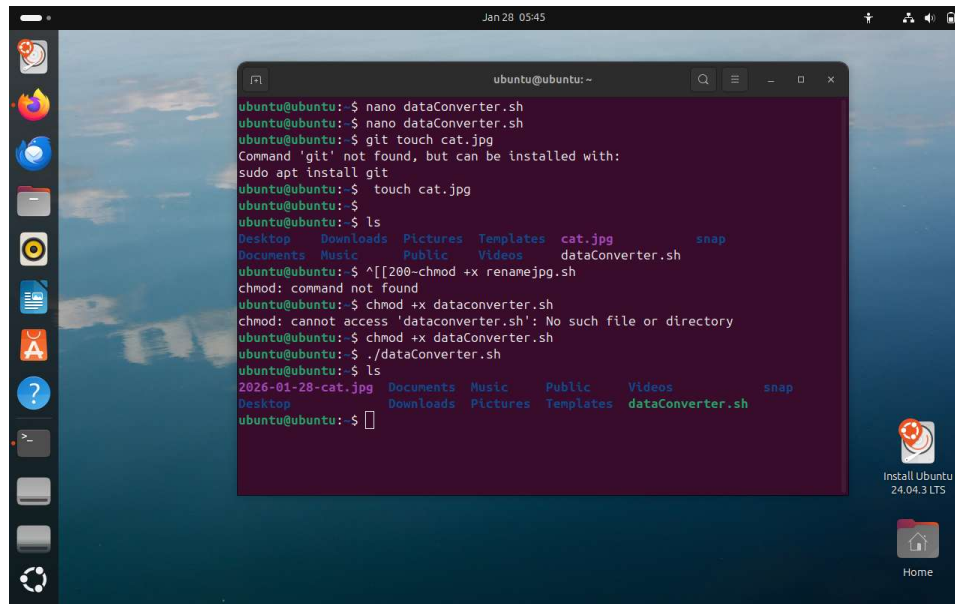
```
#!/bin/bash
```

```
date=$(date +%F)
```

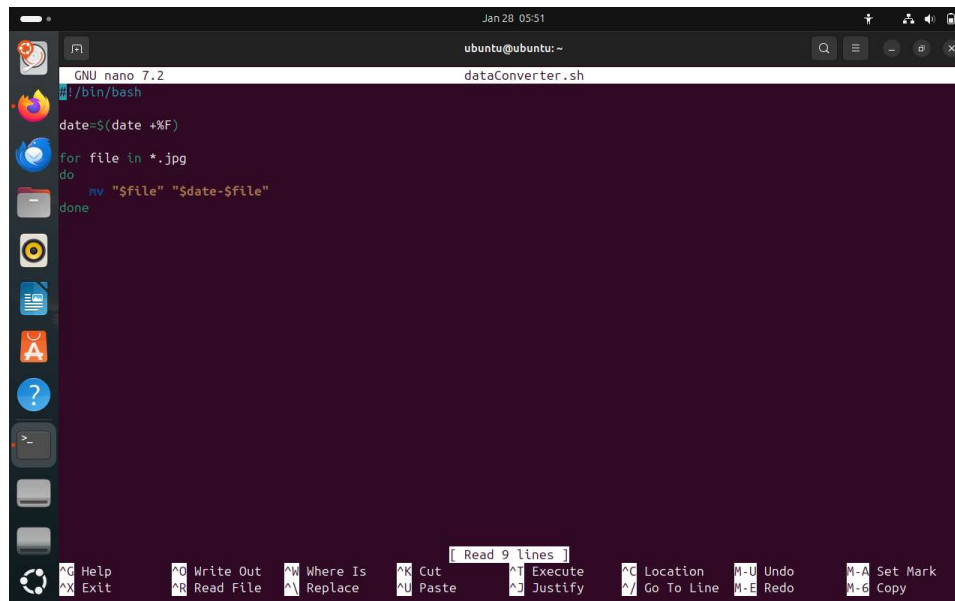
```
for file in *.jpg  
do  
    mv "$file" "$date-$file"  
done
```



```
ubuntu@ubuntu: ~  
Jan 28 05:45  
-----  
ubuntu@ubuntu: ~  
$ nano dataConverter.sh  
ubuntu@ubuntu: ~  
$ nano dataConverter.sh  
ubuntu@ubuntu: ~  
$ git touch cat.jpg  
Command 'git' not found, but can be installed with:  
sudo apt install git  
ubuntu@ubuntu: ~  
$ touch cat.jpg  
ubuntu@ubuntu: ~  
$  
ubuntu@ubuntu: ~  
$ ls  
Desktop Downloads Pictures Templates cat.jpg
```



```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ nano dataConverter.sh  
ubuntu@ubuntu:~$ nano dataConverter.sh  
ubuntu@ubuntu:~$ git touch cat.jpg  
Command 'git' not found, but can be installed with:  
sudo apt install git  
ubuntu@ubuntu:~$ touch cat.jpg  
ubuntu@ubuntu:~$  
ubuntu@ubuntu:~$ ls  
Desktop  Downloads  Pictures  Templates  cat.jpg      snap  
Documents Music      Public    Videos    dataConverter.sh  
ubuntu@ubuntu:~$ ^[[200~chmod +x renamejpg.sh  
chmod: command not found  
ubuntu@ubuntu:~$ chmod +x dataConverter.sh  
chmod: cannot access 'dataConverter.sh': No such file or directory  
ubuntu@ubuntu:~$ chmod +x dataConverter.sh  
ubuntu@ubuntu:~$ ./dataConverter.sh  
ubuntu@ubuntu:~$ ls  
2026-01-28-cat.jpg Documents Music      Public    Videos    snap  
Desktop  Downloads  Pictures  Templates  dataConverter.sh  
ubuntu@ubuntu:~$
```



```
GNU nano 7.2 dataConverter.sh  
#!/bin/bash  
  
date=$(date +%F)  
for file in *.jpg  
do  
    mv "$file" "$date-$file"  
done
```

4. Write a shell script for a program with options to calculate the following:
- a) Area of a circle
 - b) Circumference of a circle
 - c) Area of a rectangle

d) Area of a square

```
#!/bin/bash
```

```
echo "MENU"
echo "1. Area of Circle"
echo "2. Circumference of Circle"
echo "3. Area of Rectangle"
echo "4. Area of Square"
```

```
read choice
```

```
case $choice in
```

```
1)
```

```
    echo "Enter radius:"
    read r
    area=$(echo "3.14 * $r * $r" | bc)
    echo "Area of circle = $area"
    ;;
```

```
2)
```

```
    echo "Enter radius:"
    read r
    circum=$(echo "2 * 3.14 * $r" | bc)
    echo "Circumference = $circum"
    ;;
```

```
3)
```

```
    echo "Enter length:"
    read l
    echo "Enter breadth:"
    read b
    area=$(echo "$l * $b" | bc)
    echo "Area of rectangle = $area"
    ;;
```

```
4)
```

```
    echo "Enter side:"
    read s
```

area=\$(echo

```
Jan 28 05:50
ubuntu@ubuntu: ~
Install RELEASE
Downloads Pictures Templates DataConverter.sh
ubuntu@ubuntu:~$ nano area.sh
ubuntu@ubuntu:~$ sudo apt install bc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
bc is already the newest version (1.07.1-3ubuntu4).
bc set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ubuntu:~$ chmod +x area.sh
ubuntu@ubuntu:~$ ./area.sh
1. Area of Circle
2. Circumference of Circle
3. Area of Rectangle
4. Area of Square
1
Enter radius:
5
Area = 78.50
ubuntu@ubuntu:~$ ./area.sh
1. Area of Circle
2. Circumference of Circle
3. Area of Rectangle
4. Area of Square
4
Enter side:
6
Area = 36
ubuntu@ubuntu:~$ ./area.sh
1. Area of Circle
2. Circumference of Circle
3. Area of Rectangle
4. Area of Square
```

```
Jan 28 05:50
ubuntu@ubuntu: ~
GNU nano 7.2 area.sh
#!/bin/bash

echo "1. Area of Circle"
echo "2. Circumference of Circle"
echo "3. Area of Rectangle"
echo "4. Area of Square"

read ch

case $ch in
1)
echo "Enter radius:"
read r
echo "Area = $(echo "3.14*$r*$r" | bc)"
;;
2)
echo "Enter radius:"
read r
echo "Circumference = $(echo "2*3.14*$r" | bc)"
;;
3)
echo "Enter length:"
read l
echo "Enter breadth:"
read b
echo "Area = $(echo "$l*$b" | bc)"
;;
4)

```

Read 36 lines

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
Exit Read File Replace Paste Justify Go To Line M-E Redo M-C Copy